A. 1.00 GBP. each

B. $\frac{X}{12}$ GBP. each.

11 people chose A.		8	1.00
10 people chose B.	83p.	13	1.08.

$i = 1, 2, \ldots, N$

x_i flow on link i, ≥ 0.

$f_i(x_i) = a_i + b_i x_i$

link cost function, per user

a_i: free flow travel cost

b_i: "congestibility"

$\underset{\sim}{a} = \{a_i\}$, $\underset{\sim}{b} = \{b_i\}$, $\underset{\sim}{x} = \{x_i\}$.

y_j flows on routes j.

$\underset{\sim}{x} = A \underset{\sim}{y}$ — link-route incidence matrix.

$A_{ij} = 1$ if route j uses link i, 0 otherwise.

$$\underset{\sim}{A} = \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix}$$

$\underset{\sim}{d} = \begin{pmatrix} d_1 \\ d_2 \\ \vdots \\ d_p \end{pmatrix} \geq \underset{\sim}{0}$ $\quad \underset{\sim}{d} = F \underset{\sim}{y}$ OD-route incidence matrix.

$(d = y_N + y_S + y_M)$

SO. Minimise $\sum_i x_i f_i(x_i)$ subj. to $\underset{\sim}{x} = A\underset{\sim}{y}$, $\underset{\sim}{d} = F\underset{\sim}{y}$, $\underset{\sim}{y} \geq \underset{\sim}{0}$.

$\left(\sum_i x_i (a_i + b_i x_i)\right)$

$= \underset{\sim}{a}^T \underset{\sim}{x} + \underset{\sim}{x}^T \underbrace{\begin{pmatrix} b_1 & & 0 \\ & \ddots & \\ 0 & & b_n \end{pmatrix}}_{H = \operatorname{diag}(\underset{\sim}{b})} \underset{\sim}{x}$.

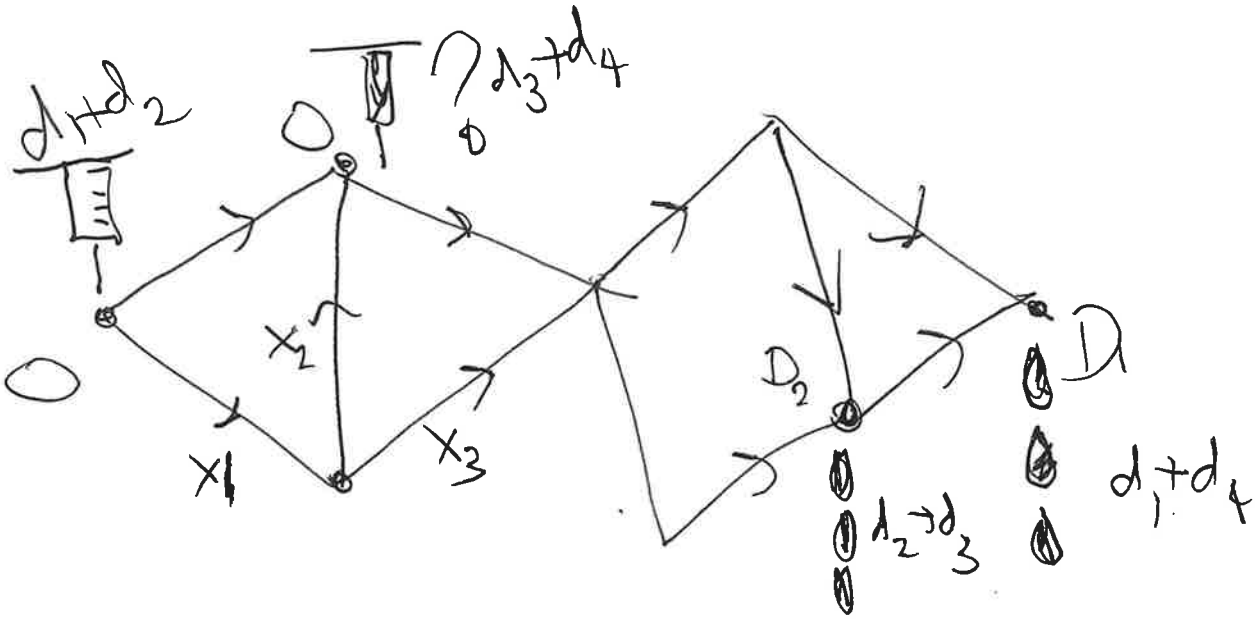
$\frac{1}{2}$ for UE

$= (\underset{\sim}{a}^T A)\underset{\sim}{y} + \underset{\sim}{y}^T (A^T H A)\underset{\sim}{y}$

Route costs?			
$\underset{\sim}{y}^*$.	$\underset{\sim}{x}^* = A\underset{\sim}{y}^*$.	$\underset{\sim}{a} + \underset{\sim}{b} \circ \underset{\sim}{x}^*$	$A^T(\underset{\sim}{a} + \underset{\sim}{b} \circ \underset{\sim}{x}^*)$

UE. Beckmann. $\sum_i \int_0^{x_i} f_i(\tilde{x})\, d\tilde{x}$

so $\sum_1 \left(a_i x_i + b_i y_i^2\right)$, UE. $\sum_i \left(a_i x_i + \frac{1}{2} b_i y_i^2\right)$



$$x_1 = x_2 + x_3$$

$$x_1 - x_2 - x_3 = 0.$$

$$\underset{\sim}{B}\underset{\sim}{x} = \underset{\sim}{s}$$

(node-link incidence matrix

$$\underset{\sim}{a}^T\underset{\sim}{x} + \frac{1}{2}\underset{\sim}{x}^T\underset{\sim}{H}\underset{\sim}{x}$$

$$\underset{\sim}{x} \geq \underset{\sim}{0}.$$

$\underset{\sim}{x}^*$. $\underset{\sim}{a} + \underset{\sim}{b} \circ \underset{\sim}{x}^*$

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q inks.

$$\underset{\sim}{X} = \begin{pmatrix} \underset{\sim}{x}^{(1)} \\ \underset{\sim}{x}^{(2)} \\ \vdots \\ \underset{\sim}{x}^{(p)} \end{pmatrix}$$

$\underset{\sim}{x}^{(1)}$ — link flows of ink 1.

" 2

$$\underset{\sim}{B}\underset{\sim}{x}^{(i)} = \underset{\sim}{s}^{(j)}$$

$$\begin{bmatrix} B & & \\ & B & \\ & & B \end{bmatrix} \underset{\sim}{X} = \begin{pmatrix} \underset{\sim}{s}^{(1)} \\ \underset{\sim}{s}^{(2)} \\ \vdots \\ \underset{\sim}{s}^{(p)} \end{pmatrix}$$

$$\underset{\sim}{X} = \underset{\sim}{G}\underset{\sim}{X}.$$